

Module 06: Learning in High School

Learning Objectives

1. Define **Learning** within the context of High School.
2. Analyze how Learning interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Learning.

Introduction

This module explores **Learning**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Learning is a critical component of the 8-part Active Inference spine, bridging the gap between Action and Communication.

Key Concepts

1. Learning as a Markov Blanket Boundary

How does Learning define the boundary between the agent and the environment?

2. Generative Models of Learning

What parameters involved in Learning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Learning drive the perception-action loop?

Applications

In High School, we see Learning manifest in: * **Specific Example 1:** Curve fitting in statistics class is mathematical learning through Active Inference: when you fit a line $y = mx + b$ to

scattered data points, you are finding the parameters (m , b) that minimize the sum of squared prediction errors -- each iteration of the least-squares algorithm updates your generative model to better predict the data, and the residuals tell you exactly how much surprise remains. *

Specific Example 2: The mathematical concept of overfitting directly maps to Active Inference's complexity term in free energy: a polynomial that passes through every data point (high-order model) has zero prediction error on training data but massive surprise on new data, while a simpler model with slightly higher training error generalizes better -- this accuracy-complexity tradeoff is the mathematical essence of why biological learning favors efficient generative models.

Conclusion

Understanding Learning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.